Employee Work Pattern Analysis

Understanding productivity through data analytics

Made with GAMMA

Project Overview

Python

Data cleaning with Pandas

MySQL

Structured data analysis

Excel

Dashboard visualization

Complete data analytics workflow analyzing employee productivity and work behavior

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Dataset Overview

50

Employee Records

13

Attributes

17

Missing Values

Key Attributes

- Work Mode (Remote/Office/Hybrid)
- Experience Years
- Training Hours
- Avg Daily Hours
- Performance Score
- Job Satisfaction

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Data Cleaning Process

01

Import & Detect

Load dataset, identify missing values

02

Replace Missing

Average age, median experience, mode satisfaction

03

Convert Types

Numeric columns to proper data types

04

Remove Duplicates

Clean dataset

05

Create Index

Productivity Index calculation

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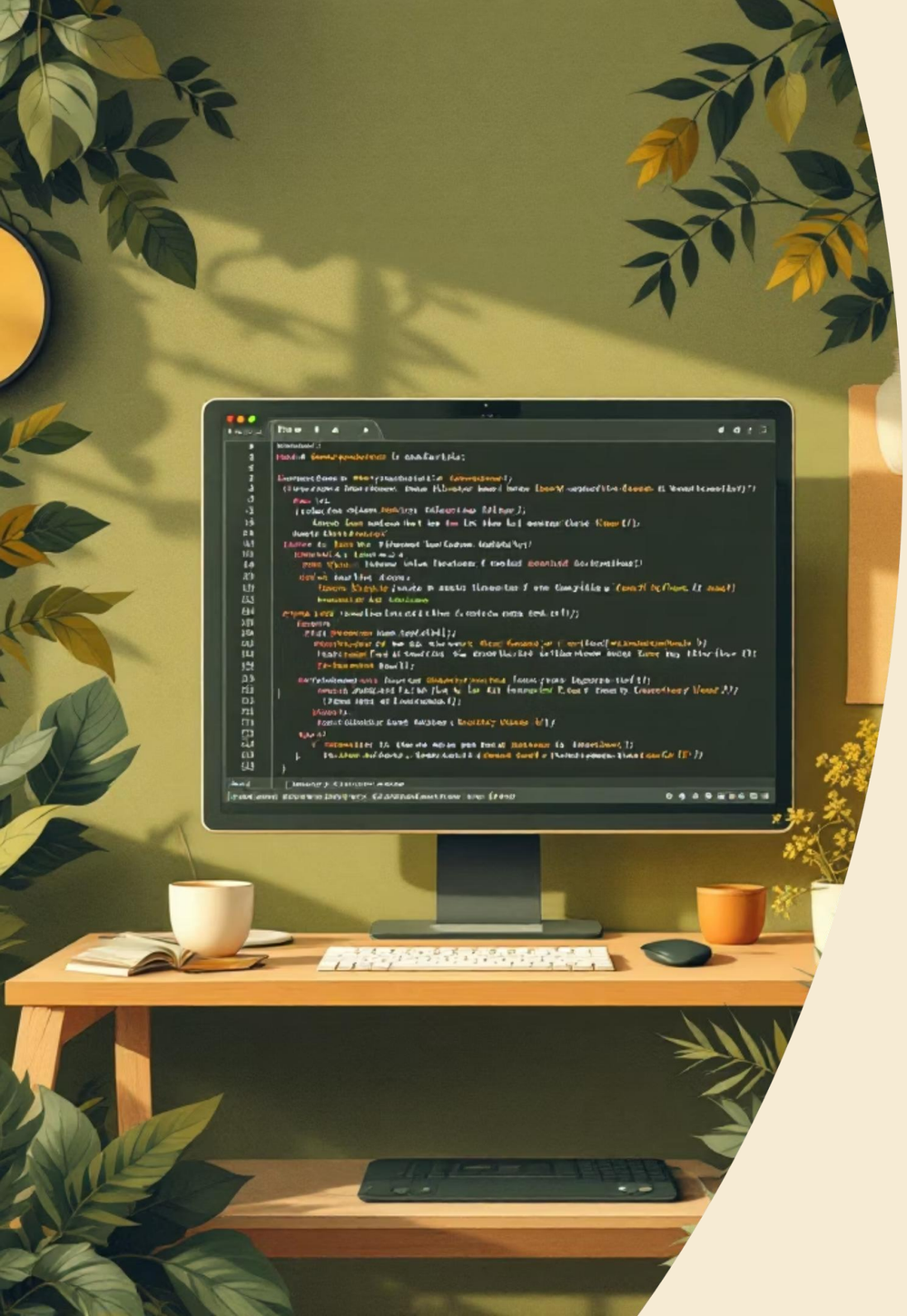

SQL Analysis Highlights

Top Performers

- Suresh: 15.04
- Prakash: 14.88
- Joseph: 14.88
- John: 13.8
- George: 13.8

Work Mode Distribution

- Office: 18 employees
- Remote: 16 employees
- Hybrid: 16 employees

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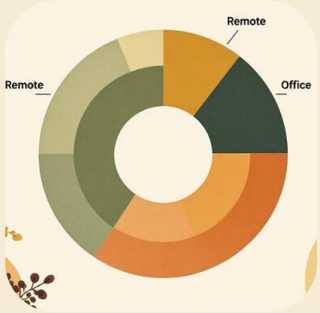
Dashboard Visualizations

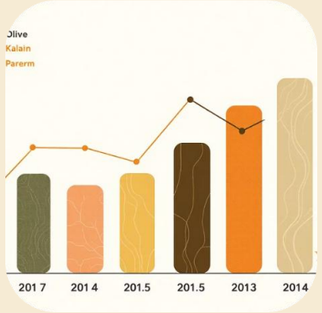
Department Distribution

IT, Marketing, Finance, HR



Work Mode Distribution

Office, Remote, Hybrid comparison



Performance by Department

IT leads at 87.4 score

Job Satisfaction

Levels 1-5 distribution

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Key Insights

Work Mode Impact

Hybrid and remote employees show similar productivity levels

Training Correlation

More training hours link to higher performance scores

Workload Balance

Balanced workloads report higher job satisfaction

Hours vs Output

Extremely long hours don't guarantee higher productivity

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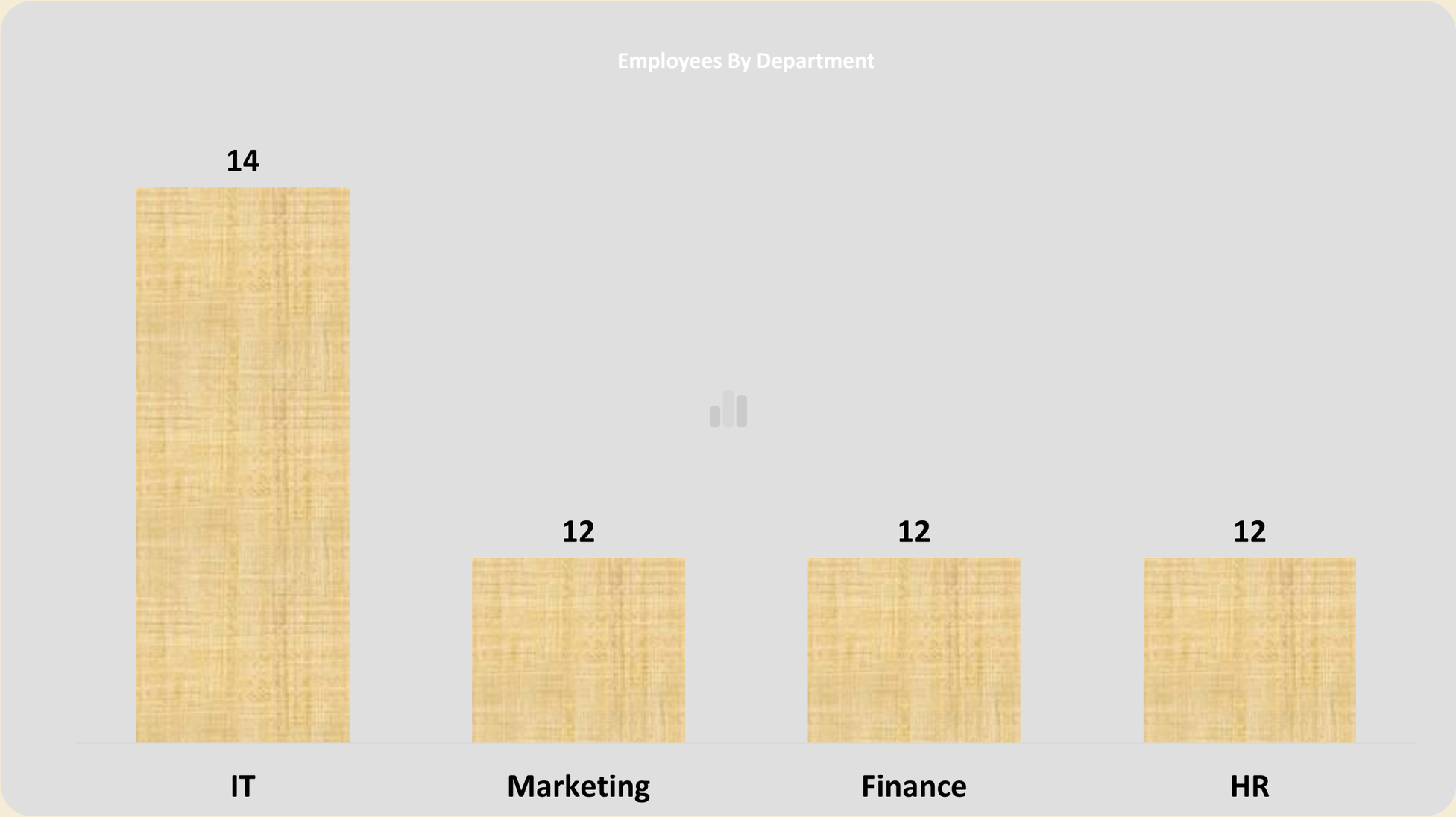
Performance by Department



IT department leads with highest average performance score

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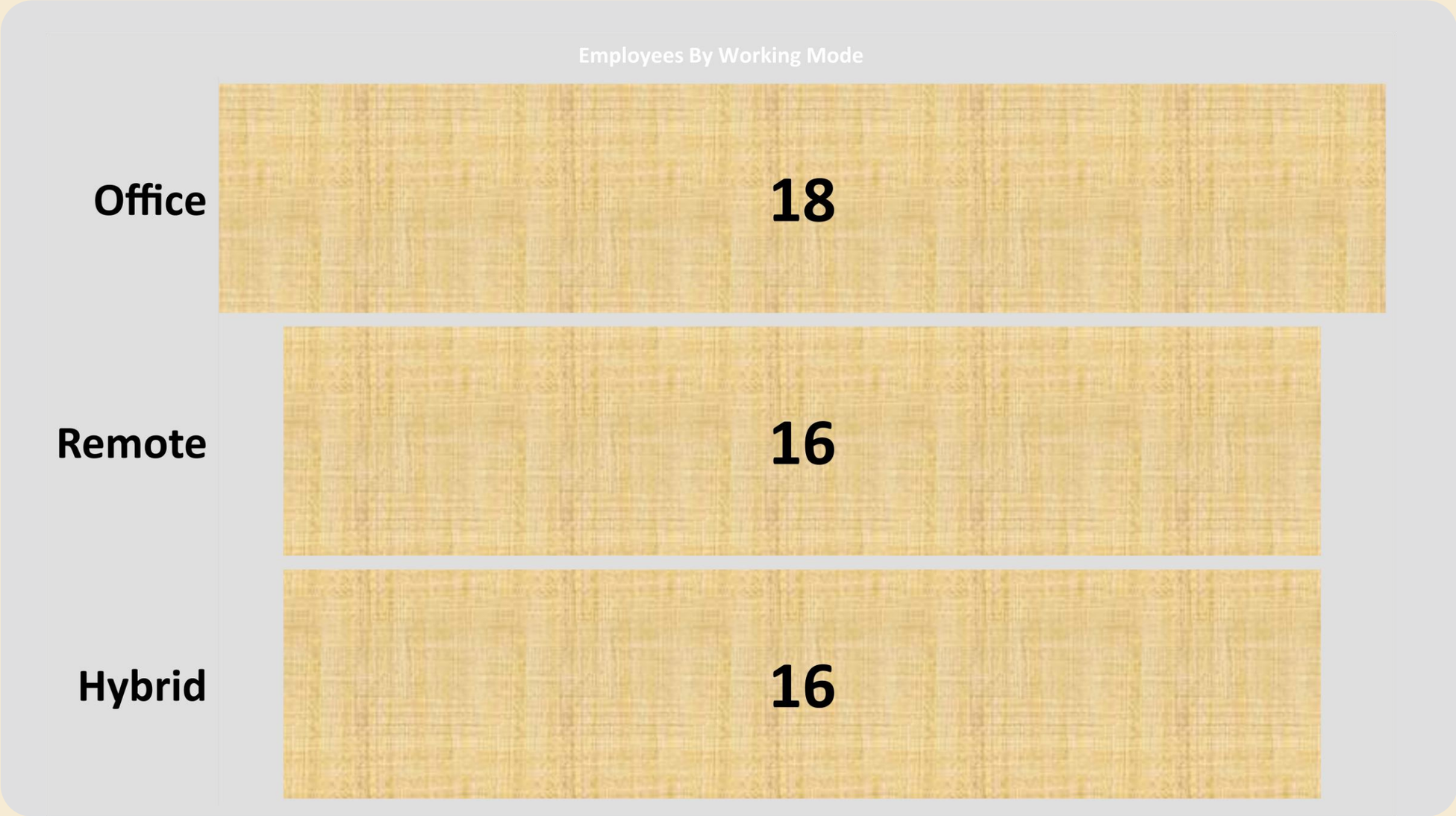
Work Mode Distribution



Office slightly leads, remote and hybrid show equal adoption

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Complete Analytics Workflow

Python

Data cleaning & preprocessing

MySQL

Structured data analysis

Excel

Visualization & dashboards

Helping organizations improve workforce management through data-driven decisions